

[MS-WPO]: IManagedObject Interface Protocol

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Content

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2.3.1 IManagedObject Interface Protocol

Summarize this article for me

.NET Framework 2.0 is an application development platform that provides several hosting interfaces that developers can use to integrate the common language runtime (CLR) into applications to provide integration between the CLR and the host's execution model. This section describes the protocols for two of these hosting interfaces: IManagedObject and IRemoteDispatch. See section 2.5 for more details on the .NET Framework protocols.

The IManagedObject Interface Protocol, specified in [[MS-IOI]] (https://learn.microsoft.com/en-us/openspecs/windows_protocols/ms-ioi/0d0efe1d-a04d-433b-b9aa-efa6cf7dc148), is a [Component Object Model \(COM\)](#) interface used by CLR to identify managed objects that are exported for interoperability with the COM and then imported back into the CLR. When the objects re-enter the CLR, they can be identified by using this interface.

The relationship of the IManagedObject interface to other protocols is described in [MS-IOI] section 1.4.

The IManagedObject interface allows COM objects to be imported as managed objects and also allows managed objects to be exported as COM objects. When COM objects are imported and used as managed objects, the CLR uses IManagedObject to determine whether an object is truly a COM object or whether it originated as a managed object.

When managed objects are exposed to COM clients as COM objects, they can implement any COM interface, but they have to at least implement `IManagedObject`.

When a COM object enters the CLR, the CLR uses the standard COM interface querying mechanism (`IUnknown::QueryInterface`) to determine whether that object implements `IManagedObject`. If the object supports `IManagedObject`, the CLR calls `IManagedObject::GetObjectIdentity`.

At CLR instantiation, the CLR creates a unique [GUID](#) to identify a specific CLR instance within a process. This GUID is formatted as a curly-braced string as defined in [\[\[MS-DTYP\]\]](#) (https://learn.microsoft.com/en-us/openspecs/windows_protocols/ms-dtyp/cca27429-5689-4a16-b2b4-9325d93e4ba2) section [2.3.4.3](#), and saved. All CLR-managed objects originating from this CLR instance return this unique identifier as the first parameter of the call to `IManagedObject::GetObjectIdentity`. This GUID is used to determine that an imported managed object originated in a particular runtime.

The CLR supports more specific levels of grouping than the process level. Objects exported from a process division can be tagged so that they return the identifier used for process division in their second parameter to `IManagedObject::GetObjectIdentity`. This identifier is also used to indicate whether the given object originated in the correct process division. If the process identifier and process division match, the last parameter of `IManagedObject::GetObjectIdentity` is a pointer to the implementation-specific representation of the managed object.

If the given object does not match the current CLR instance and process division, the CLR calls `IManagedObject::GetSerializedBuffer` to return a binary representation of a managed object, as specified by the .NET Remoting: Binary Format Data Structure [\[\[MS-NRBF\]\]](#) (https://learn.microsoft.com/en-us/openspecs/windows_protocols/ms-nrbf/75b9fe09-be15-475f-85b8-ae7b7558cfe5). The caller on the client CLR is responsible for interpreting the deserialized opaque object reference.